

Institute of Actuaries of India

Subject CT6 – Statistical Methods

September 2017 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

i)

Let h_i is the state when i^{th} horse wins the race.

The payoff matrix for various strategies are given by:

	hi (i=1 to 10)									
Strategy	1	2	3	4	5	6	7	8	9	10
Option (a)	10	20	30	40	50	60	70	80	90	100
Option (b)	5	30	45	20	75	30	105	40	45	50
Option (c)	18	4	6	72	10	12	14	16	162	20

[2]

ii)

Probability of winning various horses on rainy and other day is summarized in the table below:

	hi (i=1 to 10)									
	1	2	3	4	5	6	7	8	9	10
Rainy Day	0.067	0.133	0.067	0.133	0.067	0.133	0.067	0.133	0.067	0.133
Other Day	0.150	0.050	0.150	0.050	0.150	0.050	0.150	0.050	0.150	0.050

Probability of rain is 0.25.

Hence the weighted probability and corresponding expected payoff is given by

	hi (i=1 to 10)									
	1	2	3	4	5	6	7	8	9	10
Weight Prob	0.129	0.071	0.129	0.071	0.129	0.071	0.129	0.071	0.129	0.071
<i>Expected Payoff</i>										
Option (a)	1.292	1.417	3.875	2.833	6.458	4.250	9.042	5.667	11.625	7.083
Option (b)	0.646	2.125	5.813	1.417	9.688	2.125	13.563	2.833	5.813	3.542
Option (c)	2.325	0.283	0.775	5.100	1.292	0.850	1.808	1.133	20.925	1.417

Total expected payoff for all the strategies is given by

Strategy	Weighted Payoff
Option (a)	53.542
Option (b)	47.563
Option (c)	35.908

Hence the optimal strategy under bayes criterion is Option (a).

[7 Marks]

Solution 2:

i) Let the claims in year 0 is $X \sim \log N(\mu, \sigma^2)$

$$E(x) = e^{\mu + \frac{1}{2}\sigma^2} = 900$$

$$e^{2\mu + \sigma^2} (e^{\sigma^2} - 1) = 111300$$

$$\mu = 6.8 \text{ and } \sigma = .358$$

$$\text{The net amount paid by the reinsurer in year 1} = R_1 = \begin{cases} 0 & \text{if } X \leq 1000/1.10 \\ X - 1000 & \text{if } X > 1000/1.10 \end{cases}$$

Or

$$= \begin{cases} 0 & \text{if } X \leq 1000/1.10 \\ 1.10X - 1000 & \text{if } X > 1000/1.10 \end{cases}$$

$$\begin{aligned} E(R_1) &= \int_{1.10}^{\infty} (1.10X - 1000)f(x) dx = \int_{1.10}^{\infty} (1.10X)f(x) dx - 1000 * \int_{1.10}^{\infty} f(x) dx \\ &= 1.10 \int_{1.10}^{\infty} Xf(x) dx - 1000 * P(X > 1000/1.10) \\ &= 1.10 * e^{\mu + \frac{1}{2}\sigma^2} * [1 - \Phi((\log(1000/1.10) - 6.8 - .358^2)/.358)] - \\ &\quad 1000 * (1 - \Phi((\log(1000/1.10) - 6.8)/.358)) \\ &= 134.56 \end{aligned}$$

Similarly the net amount paid by the reinsurer in year 2 = R_2

$E(R_2)$ can be written as

$$\begin{aligned} &\int_{1.10^2}^{\infty} (1.10^2 X)f(x) dx - 1000 * \int_{1.10^2}^{\infty} f(x) dx \\ &= 1.10^2 * e^{\mu + \frac{1}{2}\sigma^2} * [1 - \Phi((\log(1000/1.10^2) - 6.8 - .358^2)/.358)] - \\ &\quad 1000 * (1 - \Phi((\log(1000/1.10^2) - 6.8)/.358)) \\ &= 195.21 \end{aligned}$$

[8]

ii) We need to find the probability that

$$P(X_n > 1000/1.10^n)$$

$$\begin{aligned} &= 1 - \Phi((\log(1000/1.10^n) - 6.8)/.358) = .8 \\ &= \Phi((\log(1000/1.10^n) - 6.8)/.358) = .2 \\ &= (\log(1000/1.10^n) - 6.8)/.358 = -.84162 \end{aligned}$$

$$n = 4.3 \text{ years}$$

[4]

[12 Marks]

Solution 3:

i) Let X is claim amount, Y is net claim payable by the insurer and Z is the reinsured part.

$$\text{So } Z = \begin{cases} X, & X \leq K \\ K + 0.6 * (X - K), & X > K \end{cases}$$

$$Y = \begin{cases} 0, & X \leq K \\ 0.4 * (X - K), & X > K \end{cases}$$

Individual claim amounts follow exponential distribution with parameter $\beta=0.02$

Now the net premium income by the insurer is given by,

$$\begin{aligned} C_{net} &= \lambda(1 + 0.34)E(X) - \lambda(1 + 0.4)E(Z) \\ &= 1.34\lambda \frac{1}{\beta} - 1.4\lambda \left[\int_0^K xf(x)dx + \int_K^\infty [K + 0.6(x - K)]f(x)dx \right] \\ &= 1.34\lambda \frac{1}{\beta} - 1.4\lambda \left[\int_0^K x\beta e^{-\beta x} dx + \int_K^\infty [K + 0.6(x - K)]\beta e^{-\beta x} dx \right] \\ &= 1.34\lambda \frac{1}{\beta} - 1.4\lambda \left[\int_0^K x\beta e^{-\beta x} dx + 0.4K \int_K^\infty \beta e^{-\beta x} dx + 0.6 \int_K^\infty x\beta e^{-\beta x} dx \right] \\ &= 1.34\lambda \frac{1}{\beta} - 1.4\lambda \left[-Ke^{-\beta K} - \frac{1}{\beta}e^{-\beta K} + \frac{1}{\beta} + 0.4Ke^{-\beta K} + 0.4(Ke^{-\beta K} + \frac{1}{\beta}e^{-\beta K}) \right] \\ &= 1.34\lambda \frac{1}{\beta} - 1.4\lambda \frac{1}{\beta} (1 - 0.4e^{-\beta K}) \\ &= \frac{\lambda}{\beta} (0.56e^{-\beta K} - 0.06) \end{aligned}$$

Net claim payable by the insurer = $\lambda E(Y)$

$$\begin{aligned} E(Y) &= \left[\int_0^K 0 \cdot f(x)dx + \int_K^\infty [0.4(x - K)]f(x)dx \right] \\ &= 0.4 \int_K^\infty (x - K)\beta e^{-\beta x} dx \\ &= \frac{0.4}{\beta} e^{-\beta K} \end{aligned}$$

The insurer will make profit if,

$$\frac{\lambda}{\beta} (0.56e^{-\beta K} - 0.06) \geq \lambda \frac{0.4}{\beta} e^{-\beta K}$$

$$\text{Or } 0.16e^{-\beta K} \geq 0.06$$

$$\text{Or } K \leq 49.04$$

[7]

ii)

Equation for adjustment co-efficient is given by

$$\lambda + C_{net} R = \lambda M_Y(R)$$

Now,

$$\begin{aligned} M_Y(R) &= E(e^{RY}) = \left[\int_0^K e^{0.4x} \beta e^{-\beta x} dx + \int_K^\infty e^{0.4(x-K)R} \beta e^{-\beta x} dx \right] \\ &= (1 - e^{-\beta K}) + \beta e^{-0.4KR} \int_K^\infty e^{-(\beta - 0.4R)x} dx \\ &= (1 - e^{-\beta K}) + \beta e^{-0.4KR} \frac{e^{-(\beta - 0.4R)K}}{\beta - 0.4R} \\ &= (1 - e^{-\beta K}) + \beta \frac{e^{-\beta K}}{\beta - 0.4R} = 1 + e^{-\beta K} \frac{0.4R}{\beta - 0.4R} \end{aligned}$$

So,

$$\lambda + \frac{\lambda}{\beta} (0.56e^{-\beta K} - 0.06)R = \lambda \left(1 + e^{-\beta K} \frac{0.4R}{\beta - 0.4R} \right)$$

$$\text{Or, } 1 + \frac{1}{\beta} (0.56e^{-\beta K} - 0.06)R = 1 + e^{-\beta K} \frac{0.4R}{\beta - 0.4R}$$

$$\text{Or, } (0.56 - 0.06e^{-\beta K})R(\beta - 0.4R) - 0.4R\beta = 0$$

$$\text{Or, } (\beta - 0.4R) = \frac{0.4\beta}{0.56 - 0.06e^{-\beta K}}, \text{ assuming } R \text{ is not equal to zero}$$

Now putting $K = 40$ and $\beta = 0.02$ we get

$$R = 0.012479$$

[6]

iii)

Aggregate claim follows compound Poisson distribution.

$$\text{Net total claim payable under part (ii)} = \lambda E(Y) = \frac{0.4}{\beta} e^{-\beta K} \lambda$$

Under Part (ii), $\beta = 0.02$, $K = 40$, $\lambda = 20$; individual mean claim amount = $1/0.02 = 50$

Hence net outgo for the insurer = 179.73

If individual claim amount in next year decreased by 30%, then revised mean claim amount = $50 \times (1 - 30\%) = 35$ and hence corresponding $\beta = 1/35 = 0.028571$

Hence the net outgo for the insurer for next year = 89.29

Total reduction in net claim is 90.44 (reduction of 50.32%)

Comment:

1. Individual claim has reduced by 30%, however net claim for the insurer has reduced by 50.32% since the retention level (K) is unchanged. As the reinsurer retains the initial claims hence higher proportion of claims has been passed on to the reinsurer.
2. Under the revised experience, the insurer will generate profit if the the retention level follow:
 $0.16e^{-\beta K} \geq 0.15$ or $K \leq 34.33$
 However current retention level is 40. Hence the insurer will make loss if the retention level is lowered below 34.33.

[5]

iv)

Under part (ii),

Mean of aggregate claim amount = $\lambda/\beta = 1,000$ Variance of aggregate claim amount = $\lambda[\text{Var}(X) + (E(X))^2] = 1,00,000$

$$\text{Net premium income for the insurer} = \frac{\lambda}{\beta} (0.56e^{-\beta K} - 0.06) = 191.62$$

Now net claim paid by the insurer = 179.73

Net claim paid by the reinsurer = $1000 - 179.73 = 820.27$ Variance of net claim paid by the insurer = $(179.73/1000) * 1,00,000 = 17,973.16$

Under part (iii),

Mean of aggregate claim amount = $\lambda/\beta = 700$ Variance of aggregate claim amount = $\lambda[\text{Var}(X) + (E(X))^2] = 49,000$

$$\text{Net premium income for the insurer} = \frac{\lambda}{\beta} (0.56e^{-\beta K} - 0.06) = 83.01$$

Now net claim paid by the insurer = 89.29

Net claim paid by the reinsurer = $1000 - 179.73 = 610.71$ Variance of net claim paid by the insurer = $(89.29/700) * 49,000 = 6,250.57$ Using normal approximation for $\psi_1(500,1)$, we getFor part (ii), $\psi_1(500,1) = P(150+191.62-N(179.73,17973.16)<0) = P(Z > 1.21) = 0.1136$ For part (iii), $\psi_1(500,1) = P(150+83.01-N(89.29,6250.57)<0) = P(Z > 1.82) = 0.0345$

[4]

[22 Marks]**Solution 4:**

i)

$$E[m(\theta)] = \bar{X} = \frac{\sum_{j=1}^{10n} N(P_{ij} X_{ij})}{\sum_{j=1}^{10n} N(P_{ij})} = 289000/10450 = 27.65$$

$$E[s^2(\theta)] = N^{-1} \sum_{i=1}^N 1/(n-1) \sum P_{ij}(X_{ij} - \bar{X}_i)^2$$

$$= 1/3 * 1/2 * 231206.8 = 38534.47$$

$$P^* = 1/(Nn-1) * \sum_{i=1}^N \bar{P}_i \{ 1 - \bar{P}_i / \sum_{i=1}^N \bar{P}_i \}$$

$$= 1/8 * 5745.45 = 718.18$$

$$V[m(\theta)] = 1/P^* \left\{ \frac{1}{Nn-1} \sum P_{ij}(X_{ij} - \bar{X})^2 - \frac{1}{n-1} \sum P_{ij}(X_{ij} - \bar{X}_i)^2 \right\}$$

$$= 1/718.18 * \left\{ \frac{1}{Nn-1} \sum P_{ij}(X_{ij} - \bar{X})^2 - E[s^2(\theta)] \right\}$$

$$= 1/718.18 * [74199.22 - 38534.47] = 49.65$$

For reinsurer A

$$\bar{Z} = 2900 / (2900 + 38534.47 / 49.65) = .788$$

Similarly for reinsurer B

$$\bar{Z} = 6250 / (6250 + 38534.47 / 49.65) = .889$$

EBCT expected claim payout per unit for reinsurer A is

$$= 20.68 * .788 + .211 * 27.65 = 22.16; \text{ expected amount} = 22.16 * 1200 = 26.59m$$

For reinsurer B it is

$$= 32 * .889 + .110 * 27.65 = 31.52; \text{ expected amount} = 31.52 * 2200 = 69.34m \quad [8]$$

ii)

The credibility factor is an increasing function of n. The more the data there are from the risk itself, the more it will be relied on when the credibility estimate of premium or number of claims is calculated.

The credibility factor is an increasing function of $V[m(\theta)]$. Higher the value of $V[m(\theta)]$, relative to $E[s^2(\theta)]$, more variability there is between different risks in the collective and hence the less likely it is that the other risks in the collective will resemble the risk that is of interest. The less reliance will be placed on the data from these other risks.

P_j 's specify the relative amount of weightings to be put on the claims for each year. A uniform increase applied to all weights is not expected to affect the credibility factor. [4]

[12 Marks]

Solution 5:

i)

Model	Linear Predictor	Number of parameters	Scaled Deviance
VS	$a + bx$	2	500
$\text{Log}(AA) + AA^2$	$\alpha + \beta \log(y) + \lambda y^2$	3	620

VS + ST	$a_i^{\wedge} + bx$	13	475
AA + exp(AS)	$\alpha_i + \beta y$	16	540
VS * FL + ST	$a_i^{\wedge\wedge} + b_i^{\wedge\wedge}x + B_j$	31	420
AA + AS + AA*AS	$\alpha_i' + \beta_i' y$	30	450
VS * ST * FL	$a_i^{\wedge\wedge\wedge} + b_i^{\wedge\wedge\wedge}x$	240	350

VS: $a+bx$ 2 parameters
VS+ST: $a+bx + Si = a_i^{\wedge} + bx$ $2+(12-1) = 13$ parameters
VS * FL + ST: $(a+bx)*Fi + Si = a_i^{\wedge\wedge} + b_i^{\wedge\wedge}x + B_j$ $2*10+(12-1)=31$ paramters
VS * ST * FL: $(a+bx)* Si * Fi = a_i^{\wedge\wedge\wedge} + b_i^{\wedge\wedge\wedge}x$ $2*12*10 = 240$ parameters

AA: $\alpha + \beta y$ 2 parameters
Log(AA) + AA²: $\alpha + \beta \log(y) + \lambda y^2$ 3 parameters
AA + exp(AS): $\alpha + \beta y + \exp(A_i) = \alpha_i + \beta y$ $2+(15-1) = 16$ parameters
AA + AS + AA*AS: $(\alpha + \beta y) * A_i = \alpha_i' + \beta_i' y$ $2*15 = 30$ parameters

[5]

ii)

*Comparing VS and 'VS + ST'*Difference in Scaled deviance = $500 - 475 = 25$ Now, $\chi^2_{(13-2)} = \chi^2_{11} = 19.68$, at 5% critical level; which is lower than the scaled deviance.

Hence 'VS + ST' is significant improvement over VS.

*Comparing 'VS + ST' and 'VS * FL + ST'*Difference in Scaled deviance = $475 - 420 = 55$ Now, $\chi^2_{(31-13)} = \chi^2_{18} = 28.87$, at 5% critical level; which is lower than the scaled deviance.

Hence 'VS * FL + ST' is significant improvement over 'VS + ST'.

*Comparing 'VS * FL + ST' and 'VS * ST * FL'*Difference in Scaled deviance = $420 - 350 = 70$ Now, $\chi^2_{(240-31)} = \chi^2_{209} > 124.3$, at 5% critical level; which is higher than the scaled deviance.

Hence 'VS * ST * FL' is not significant improvement over 'VS * FL + ST'.

Hence optimal model for value of ship (VS) is 'VS * FL + ST'

*Comparing 'Log(AA) + AA²' and 'AA + exp(AS)'*Difference in Scaled deviance = $620 - 540 = 80$ Now, $\chi^2_{(16-3)} = \chi^2_{13} = 22.36$, at 5% critical level; which is lower than the scaled deviance.Hence 'AA + exp(AS)' is significant improvement over 'Log(AA) + AA²'.

Comparing 'AA + exp(AS)' and 'AA + AS + AA*AS'

Difference in Scaled deviance = 540 - 450 = 90

Now, $\chi^2_{(30-16)} = \chi^2_{14} = 23.68$, at 5% critical level; which is lower than the scaled deviance.

Hence 'AA + AS + AA*AS' is significant improvement over 'AA + exp(AS)'.

Hence optimal model for average age of ship worker (AA) is 'AA + AS + AA*AS'. [5]

iii)

We should check the below points before deciding on the final model to be considered:

- To ensure good fit of the data we should check the residuals
- The significance of the parameters of the proposed model to ensure that all the estimated parameters are significantly different from zero
- Other models involving the various factors
- Order of factors may have impact on the improvement parameters
- Level of critical value

[2]

[12 Marks]

Solution 6:

i) Purposes of a practical time series analysis:

- description of the data
- construction of a model which fits the data
- forecasting future values of the process
- deciding whether the process is out of control, requiring action
- for vector time series, investigating connections between two or more observed processes with the aim of using values of some of the processes to predict those of the others

[2]

ii)

Univariate time series

A univariate time series is a sequence of observations $\{X_t\}$ recorded at regular intervals. The state space is continuous but the time set is discrete. Such series may follow a pattern to some extent, for example possessing a trend or seasonal component, as well as having random factors.

Invertibility

A time series process X is invertible if we can write the white noise term e_t as a convergent sum of the X terms. It can be shown that this is equivalent to saying that the roots of the characteristic polynomial of the e terms are all greater than 1 in magnitude. Invertibility is a desirable characteristic since it enables us to calculate the residual terms and hence analyse the goodness of fit of a particular model.

Markov

A time series process X has the Markov property if:

$$P[X_t = a \mid X_{s1} = x_1, \dots, X_{sn} = x_n, X_s = x] = P[X_t = a \mid X_s = x]$$

for all times $s_1 < s_2 < \dots < s_n < s < t$ and all states $a, x_1, \dots, x_n$ of S .

In other words we can predict the future state (at time t) from the current state (at time s) alone.

[3]

iii)

Following two methods can be used to remove the seasonal variation

1. Seasonal differencing

Half yearly variations mean the period is two half years ie $h_t = h_{t+2}$

Hence we can subtract the value from 2 half years age:

$$\nabla_2 S_t = S_t - S_{t-2}$$

2. Method of moving average

Half yearly variations mean the period is two half years ie $h_t = h_{t+2}$

We can find a symmetrical average of 2 terms about S_t

$$\frac{1}{2} \left(\frac{1}{2} S_{t-1} + S_t + \frac{1}{2} S_{t+1} \right)$$

Following two methods can be used to remove any linear trend

1. Least squares trend removal

We can estimate α and β using least square regression method,

Determine α and β which minimizes $\sum x_t^2 = \sum (y_t^2 - \alpha - \beta t)^2$

Subtract the regression line from the observed values $S_t - \alpha - \beta t$

2. Differencing

Subtract the previous observed values

$$\nabla S_t = S_t - S_{t-1} \quad [2]$$

iv) a)

$$Y_t - 0.6Y_{t-1} = 0.5 + 0.2e_t + 0.7e_{t-1} + 0.3e_{t-2}$$

$$\text{Or, } (1-0.6B) Y_t = 0.5 + 0.2e_t + 0.7e_{t-1} + 0.3e_{t-2}$$

$$\text{Or, } Y_t = (1-0.6B)^{-1} (0.5 + 0.2e_t + 0.7e_{t-1} + 0.3e_{t-2})$$

$$\text{Or, } Y_t = \sum_{i=0}^{\infty} 0.6^i B^i (0.5 + 0.2e_t + 0.7e_{t-1} + 0.3e_{t-2})$$

$$= \frac{0.5}{1-0.6} + 0.2 \sum_{i=0}^{\infty} 0.6^i e_{t-i} + 0.7 \sum_{i=0}^{\infty} 0.6^i e_{t-1-i} + 0.3 \sum_{i=0}^{\infty} 0.6^i e_{t-2-i}$$

$$= 1.25 + 0.2(1e_t + 0.6e_{t-1}) + 0.7(e_{t-1}) + (0.3 + 0.7 * 0.6 + 0.2 * 0.6^2) \sum_{i=2}^{\infty} 0.6^{i-2} e_{t-i}$$

$$= 1.25 + 0.2e_t + 0.19e_{t-1} + 1.192 \sum_{i=2}^{\infty} 0.6^{i-2} e_{t-i}$$

b)

Now

$$\text{Cov}(Y_t, e_t) = \text{Cov}(0.6Y_{t-1} + 0.5 + 0.2e_t + 0.7e_{t-1} + 0.3e_{t-2}, e_t) = 0.2\sigma^2$$

$$\text{Cov}(Y_t, e_{t-1}) = \text{Cov}(0.6Y_{t-1} + 0.5 + 0.2e_t + 0.7e_{t-1} + 0.3e_{t-2}, e_{t-1}) = 0.82\sigma^2$$

$$\text{Cov}(Y_t, e_{t-2}) = \text{Cov}(0.6Y_{t-1} + 0.5 + 0.2e_t + 0.7e_{t-1} + 0.3e_{t-2}, e_{t-2}) = 0.792\sigma^2$$

$$\text{Cov}(Y_{t-1}, e_{t-2}) = \text{Cov}(0.6Y_{t-2} + 0.5 + 0.2e_{t-1} + 0.7e_{t-2} + 0.3e_{t-3}, e_{t-2}) = 0.82\sigma^2$$

$$\begin{aligned} \text{Now, } Y_0 &= \text{Cov}(Y_t, Y_t) = \text{Cov}(Y_t, 0.6Y_{t-1} + 0.5 + 0.2e_t + 0.7e_{t-1} + 0.3e_{t-2}) \\ &= 0.6Y_1 + 0.2 \cdot 0.2\sigma^2 + 0.7 \cdot 0.82\sigma^2 + 0.3 \cdot 0.792\sigma^2 = 0.6Y_1 + 0.8516\sigma^2 \text{ -----(a)} \end{aligned}$$

$$\begin{aligned} Y_1 &= \text{Cov}(Y_t, Y_{t-1}) = \text{Cov}(Y_{t-1}, 0.6Y_{t-1} + 0.5 + 0.2e_t + 0.7e_{t-1} + 0.3e_{t-2}) \\ &= 0.6Y_0 + 0.7 \cdot 0.2\sigma^2 + 0.3 \cdot 0.82\sigma^2 = 0.6Y_0 + 0.386\sigma^2 \text{ -----(b)} \end{aligned}$$

$$Y_2 = \text{Cov}(Y_t, Y_{t-2}) = \text{Cov}(Y_{t-2}, 0.6Y_{t-1} + 0.5 + 0.2e_t + 0.7e_{t-1} + 0.3e_{t-2}) = 0.6Y_1 + 0.06\sigma^2$$

Now from (a) and (b),

$$Y_0 = 1.6925\sigma^2$$

$$Y_1 = 1.4015\sigma^2$$

$$Y_2 = 0.9009\sigma^2$$

$$\text{And } Y_k = 0.6Y_{k-1} \text{ for } k \geq 3$$

$$\text{Hence, } \rho_0 = 1, \rho_1 = 0.8281, \rho_2 = 0.5323, \rho_k = 0.6\rho_{k-1} \text{ for } k \geq 3$$

[4]

[14 Marks]

Solution 7:The likelihood of observing a sample of claims $X_1, X_2, \dots, X_n$ is

$$\begin{aligned} \prod_{i=1}^n f(x_i, \gamma) &= \prod_{i=1}^n c\gamma x_i^{\gamma-1} \exp(-cx_i^\gamma) \\ &\sim c^n \exp(-c \sum_{i=1}^n x_i^\gamma) \end{aligned}$$

The prior distribution of C is given by $f(c) = \lambda \exp(-c\lambda)$

So the pdf of the posterior distribution is given by

$$f(c|x) \sim \text{prior} \times \text{likelihood} = c^n \exp(-c \sum_{i=1}^n x_i^\gamma) * \lambda \exp(-c\lambda)$$

$$\sim C^n \exp(-C(\sum_{i=1}^n x_i^\gamma + \lambda))$$

This has the form of gamma($n+1, \sum X_i + \lambda$) distribution.

[3 Marks]

Solution 8:

i)

The characteristic equation is given by

$$= (1 - 2/5\lambda) = 0$$

$$\Rightarrow \lambda = 5/2 > 1. \text{ Hence the process is weakly stationary as its roots are greater than 1.}$$

[2]

ii) ACF for lags 0,1 and 2

$$\text{Cov}(e_t, X_t) = \text{Var}(e_t) = 6^2$$

$$\text{Cov}(e_{t-1}, X_t) = 2/5 \text{cov}(e_{t-1}, X_{t-1}) - 1/4 \text{Var}(e_{t-1}) = (2/5 - 1/4) 6^2 = 3/20 6^2$$

$$\text{Cov}(e_{t-2}, X_t) = 1/5 6^2$$

Taking covariance of the defining equation with X_t gives;

$$Y_0 = 2/5 Y_1 + 6^2 - 1/4 * 3/20 6^2 + 1/5 * 1/5 6^2 = 2/5 Y_1 + 6^2$$

$$Y_1 = 2/5 Y_0 - 1/4 6^2 + 1/5 * 3/20 6^2$$

$$Y_2 = 2/5 Y_1 + 1/5 6^2$$

Solving the equations simultaneously we get

$$Y_0 = 38/35 6^2$$

$$Y_1 = 3/14 6^2$$

$$Y_2 = 2/7 6^2$$

Thus; $\rho_0 = 1$, $\rho_1 = 15/38$, $\rho_2 = 5/19$

[6]

iii) For this process we can see that the value of X_t depends upon the value of e_{t-2} which may tell something about the value of X_{t-2} . In turn we see that the value of X_{t-2} as well as the value of X_{t-1} will influence the distribution of X_t . Hence the process is not markov.

If we express the process as a multivariate process with two components as $\begin{pmatrix} X_t \\ X_{t-1} \end{pmatrix}$, $t = 1 \text{ to } \infty$ then the relationship:

$$\begin{pmatrix} X_t \\ X_{t-1} \end{pmatrix} = \begin{pmatrix} 2/5 & 0 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} X_{t-1} \\ X_{t-2} \end{pmatrix} + \begin{pmatrix} 1 & -1/4 & 1/5 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} e_t \\ e_{t-1} \\ e_{t-2} \end{pmatrix} + \begin{pmatrix} 4 \\ 0 \end{pmatrix}$$

Which can be written as $X_t = 4 + (2/5)X_{t-1} + e_t - (1/4)e_{t-1} + (1/5)e_{t-2}$ which is the given equation and $X_{t-1} = X_{t-1}$ which is true.

This expression has multivariate markov property because knowing the value of $\begin{pmatrix} X_{t-2} \\ X_{t-3} \end{pmatrix}$ will not give any extra information for the value of $\begin{pmatrix} X_t \\ X_{t-1} \end{pmatrix}$ given the value of $\begin{pmatrix} X_{t-1} \\ X_{t-2} \end{pmatrix}$.

[4]

[12 Marks]

Solution 9:

i)

EBCT Model 1 can be regarded as a generalisation of the normal/normal model. Similarities between EBCT Model 1 and the normal/normal model are summarized below:

1. The role of the parameter θ is the same for both models: it characterises the underlying distributions of the processes being modelled, eg the aggregate claim distribution for each year of business.
2. Assumptions concerning the unconditional distribution of the X_j 's are the same: they are identically distributed in each case.

- Assumptions concerning the (conditional) distribution of the X_j 's given θ are the same: they are conditionally independent in each case.

Differences between EBCT Model 1 and the normal/normal model are summarized below:

- $E(X_j|q)$ is some function of θ , $m(\theta)$, for EBCT Model 1 but is simply θ for the normal/normal model. Hence:
 $\text{var}[m(\theta)] \{EBCT\}$ corresponds to $\sigma_2^2 (= \text{var}(\theta)) \{normal/normal\}$.
- $\text{var}(X_j | \theta)$ is a function of θ , $s^2(\theta)$, for EBCT Model 1 but is a constant, σ_1^2 , for the normal/normal model. Hence:
 $E(s^2(\theta))$ corresponds to $\sigma_1^2 (= \text{var}(X_j | \theta))$.
- The normal/normal model makes very precise distributional assumptions about both X_j given θ , which is $N(\theta, \sigma_1^2)$, and θ , which is $N(\mu, \sigma_2^2)$.
 EBCT Model 1 makes no such distributional assumptions.
- The risk parameter, θ , is a real number for the normal/normal model but could be a more general quantity for EBCT Model 1.

[4]

ii)

Y_{ij} represents the number of claims (or aggregate claim amount) for risk i ($i = 1, \dots, N$) in year j ($j = 1, \dots, n$).

P_{ij} represents the corresponding risk volume (eg number of policies or premium income). The P_{ij} 's are assumed to be known.

$$X_{ij} = Y_{ij} / P_{ij}$$

Assumptions: For each risk i , the distribution of X_{ij} depends on a parameter θ_i whose value is the same for each j but is unknown.

$\{X_{ij} | \theta_i\}$ are independent random variables.

$\{\theta_i\}$ are i.i.d random variables.

For $i \neq k$, the pairs (X_{ij}, θ_i) and (X_{km}, θ_k) are independent random variables.

There exist functions $m(\cdot)$ and $s^2(\cdot)$ such that:

$$m(\theta_i) = E(X_{ij} | \theta_i) \quad s^2(\theta_i) = P_{ij} \text{var}(X_{ij} | \theta_i)$$

[2]

[6 Marks]
